

Capacitance Level Transmitter for Fuel Applications

Model: CPLT1200F

PRODUCT DESCRIPTION:



The CPLT-1200F Capacitance Level Transmitter is specially designed for continuous level measurement of fuel and hydrocarbon liquids such as diesel, petrol, ethanol, solvents and oils. It operates on the principle of capacitance variation due to changes in liquid level and provides a reliable 4–20 mA output proportional to the measured level.

The transmitter is provided with a PTFE-insulated rigid probe with grounding tube, offering excellent insulation and stable measurement performance for fuel-level applications. The probe is suitable for tank depths up to 3 m and is available with different mounting arrangements to meet specific installation requirements.

The CPLT-1200F features easy field calibration for 4 mA (empty) and 20 mA (full) levels and is suitable for reliable level monitoring in diesel, petrol, ethanol, solvents and oil storage tanks.

SALIENT FEATURES:

- Specially designed for continuous fuel level measurement
- Suitable for petrol, diesel, furnace oil and other hydrocarbon-based fuels
- Capacitance-based level measurement technology
- Standard 4–20 mA output
- Rigid probe with grounding tube for stable and reliable fuel measurement
- PTFE-insulated probe for excellent electrical insulation
- Process temperature up to 150°C
- Measurement ranges up to 2 m with rigid probe
- Easy 4 mA / 20 mA Push-Button field calibration
- High-resolution and accurate level measurement
- No moving parts – low maintenance
- Robust weatherproof / flameproof enclosure
- Available with various process connections and mounting arrangements
- Optional digital level display and relay control
- Suitable for PLC, DCS, SCADA and automation systems
- Customised probe length and process connection options available

OPERATING PRINCIPLE:

The CPLT-1200F operates on the principle of capacitance variation caused by changes in the fuel level. The insulated sensing probe and the grounding tube/tank wall form a capacitance system. As the fuel level rises or falls, the dielectric medium surrounding the probe changes, resulting in a corresponding change in capacitance.

The electronic circuit continuously measures this capacitance change and converts it into a 4–20 mA output signal proportional to the fuel level. 4 mA represents the minimum/empty level and 20 mA represents the maximum/full level.

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PROBE CONSTRUCTION:

The CPLT-1200F is designed with a rigid, PTFE-insulated capacitance sensing probe specifically developed for fuel-level measurement applications. The PTFE insulation provides excellent electrical isolation between the sensing electrode and the process liquid, ensuring stable and reliable capacitance measurement.

A dedicated grounding tube is provided as a reference electrode to maintain consistent sensing performance, particularly in fuel and other non-conductive liquid applications. The robust probe construction is suitable for tank depths up to 3 m and can be supplied with different probe lengths and mounting arrangements to meet specific application requirements.

TECHNICAL SPECIFICATIONS : CPLT-1200F

Parameter	Specification
Type	Capacitance Level Transmitter
Supply	24 V DC, 4-Wire
Output	4–20 mA
Permissible Load	500 Ω
Measurement Range – Rigid Probe	200 to 2000 mm
Accuracy	$\pm 0.5\%$ of Full Scale
Max. Pressure	Up to 5 kg/cm ²
Operating Temperature	25 to 120°C
Probe Type	Rigid
Probe Material	SS304/SS316/MS/ Customised
Probe Insulation	PTFE Insulated
Process Connection	1" / 1.5" BSP / NPT / Flanged
Flange Size	As per Requirement
Flange Rating	ANSI 150# / 300# or as specified
Cable Entry	1/2" / 3/4" ET / NPT/ PG 13.5MM
Calibration	Push-Button Calibration
Electronics	Integral Electronics
Enclosure	Die-Cast Aluminium, Flameproof, IP66
Display	Optional Process Indicator
Mounting	Top Mounted

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Parameter	Specification
Application	Fuel, diesel, petrol, ethanol, solvents Compatible Liquids
Customization	Probe Length, Process Connection, Flange & Cable Entry as per Requirement

ELECTRICAL CONNECTIONS:

Supply: 24 VDC Output: 4 To 20 mA



Optional Process Indicator Display

Power supply: 230 V AC 50 Hz / 24 V DC @ 3 VA Max Specify.

Out Put: standard SPDT Relay x2 Nos, 4-20 mA Retransmission

For Sensor Supply: 24 V DC 2 250 mA

Max. Modbus.

OPTIONAL Communication Output: Programable RS 485 Modb

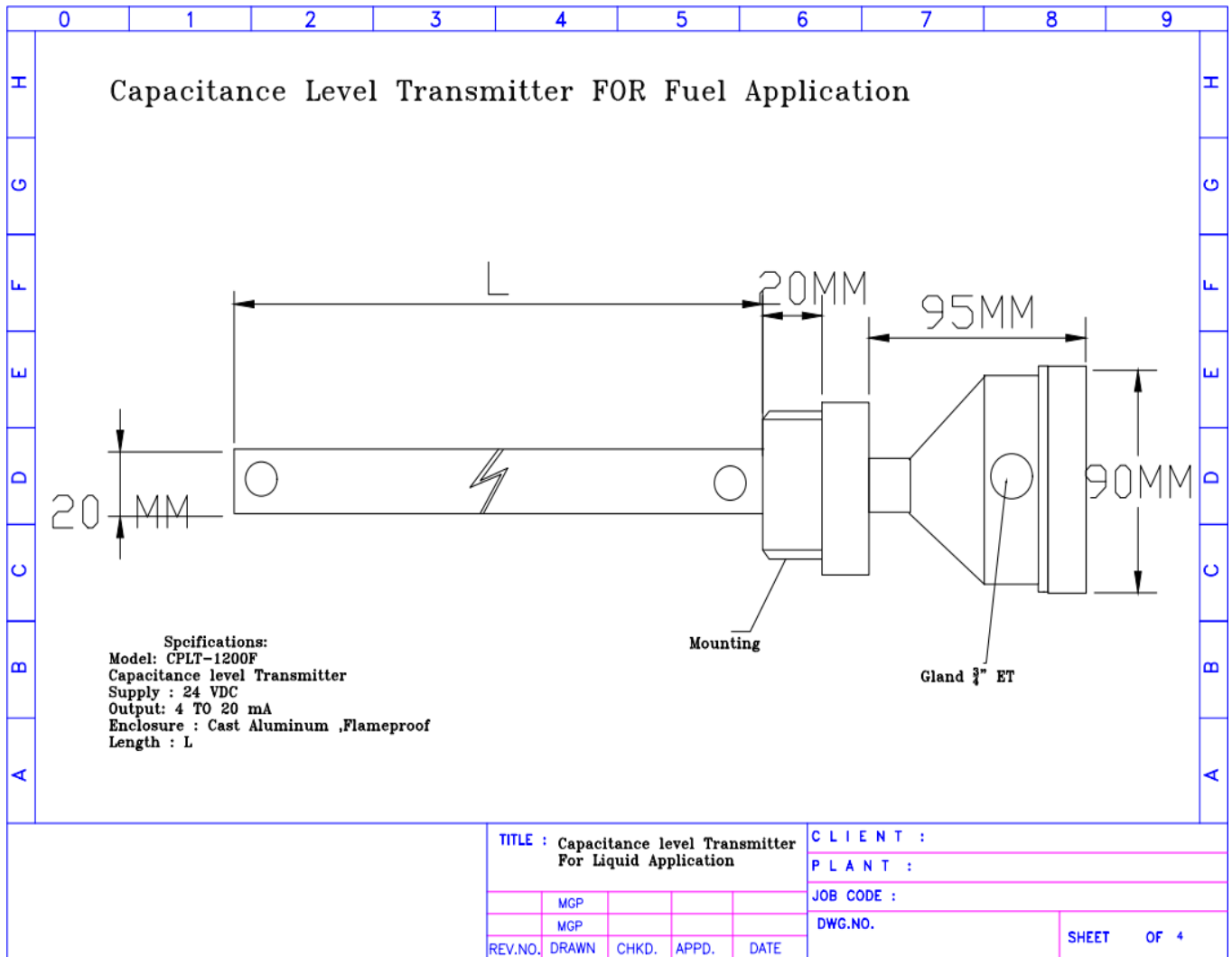
Relay output: 2 x Potential free C/O contacts rated at 6A 230 V AC 50 Hz



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Mechanical Drawing:



Manufactured By:

Levtron Instruments Pvt Ltd.

Office No 2, 1st floor, plot no 6, Pandhari Industrial estate, Shivane. Pune 411023, Contact: +91 9579430012/+91 9730042725,

E-mail: sales@levtron.in www.Levtron.in/

www.Levtroninstruments.com/www.levelsence.in